CSC 231 — Digital Logic Design

Boolean Algebra & Logic Gates

Classwork Problems with Solutions

Week 2 · Summer 2026 · Mansur Mulk, PhD.

Section 1

Postulates & Theorems

Apply Huntington's postulates to prove Boolean identities

CW 1 — Prove the Boolean Identities

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Use Huntington Postulates (P1–P6) and Theorems (T1–T8)

Prove each of the following using only the postulates and theorems of Boolean algebra. State the law used at each step.

(a) **Prove:**

$$x + xy = x$$

Hint: Use absorption / idempotency / distribution

(b) **Prove:**

$$x(x + y) = x$$

Hint: Use absorption / idempotency / distribution

(c) **Prove:**

$$x + x'y = x + y$$

Hint: Use absorption / idempotency / distribution

(d) **Prove:**

$$(x + y)(x + y') = x$$

Hint: Use absorption / idempotency / distribution

(e) **Prove:**

$$xy + xy' = x$$

Hint: Use absorption / idempotency / distribution

(f) **Prove:**

$$x'y + xy' + xy = x + y$$

Hint: Use absorption / idempotency / distribution

CW 1 — Solutions: Prove Boolean Identities

Each step cites the postulate or theorem used

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(a) **$x + xy$**

$= x \cdot 1 + xy$ [P2b]
$= x(1+y)$ [P5a]
$= x \cdot 1$ [T2a]
$= x$ [P2b] ✓

(b) **$x(x + y)$**

$= xx + xy$ [P5a]
$= x + xy$ [T1b]
$= x$ [T6a Absorption] ✓

(c) **$x + x'y$**

$= (x+x')(x+y)$ [P5b distributive]
$= 1 \cdot (x+y)$ [P6a]
$= x+y$ [P2b] ✓

(d) **$(x+y)(x+y')$**

$= x + yy'$ [P5b]
$= x + 0$ [P6b]
$= x$ [P2a] ✓

(e) **$xy + xy'$**

$= x(y+y')$ [P5a]
$= x \cdot 1$ [P6a]
$= x$ [P2b] ✓

(f) **$x'y + xy' + xy$**

$= x'y + x(y'+y)$ [P5a]
$= x'y + x \cdot 1$ [P6a]
$= x'y + x$ [P2b]
$= x + y$ [T5a: $x+x'y=x+y$] ✓

Section 2

Boolean Functions & Truth Tables

Evaluate functions, build truth tables, identify gate diagrams

CW 2 — Truth Table & Gate Diagram: $F_1 = x + yz'$

Section 2: Boolean Functions

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(a) Complete the truth table for $F_1 = x + yz'$. (b) Draw the minimal two-level AND-OR gate diagram. (c) Identify the minterms where $F_1 = 1$.

x	y	z	yz'	F_1=x+yz'	Minterm
0	0	0	0	0	—
0	0	1	0	0	—
0	1	0	1	**1**	$m_2 = x'yz'$
0	1	1	0	0	—
1	0	0	0	**1**	$m_4 = xy'z'$
1	0	1	0	**1**	$m_5 = xy'z$
1	1	0	1	**1**	$m_6 = xyz'$
1	1	1	0	**1**	$m_7 = xyz$

Gate Diagram — $F_1 = x + yz'$

Inputs: x, y, z

Level 1 (NOT): z → NOT → z'

Level 1 (AND): y, z' → AND → yz'

Level 2 (OR): x, yz' → OR → F_1

Gate count: 1 NOT + 1 AND + 1 OR = 3 gates
Gate delays: 2 levels (AND→OR)

$F_1 = 1$ for minterms: {2, 4, 5, 6, 7}
$F_1(x,y,z) = \Sigma m(2, 4, 5, 6, 7)$

CW 3 — Simplify & Implement: $F_2 = x'y'z + x'yz + xy'$

Section 2: Equivalent Logic / Algebraic Manipulation

(a) Simplify F_2 to a minimal SOP. (b) Draw both the original and simplified gate diagrams. (c) State how many gates are saved.

Original expression (3 terms, 8 literals):

$$F_2 = x'y'z + x'yz + xy'$$

✓ Solution — Step-by-Step

Step 1 — Factor out x'z from first two terms:
$x'y'z + x'yz = x'z(y' + y) = x'z \cdot 1 = x'z$

Step 2 — Write simplified form:
$F_2 = x'z + xy'$

Result: 2 terms, 4 literals (saves 4 literals!)

Gate comparison:
Original: 3 AND (3-input) + 1 OR (3-input) + 3 NOT = 7 gates
Simplified: 2 AND (2-input) + 1 OR + 2 NOT = 5 gates
Savings: 2 gates, 4 fewer literals

Gate Diagrams (Fig. 2.2)

(a) Original — $F_2 = x'y'z + x'yz + xy'$:
NOT(x) → x'   NOT(y) → y'
AND(x', y', z) → x'y'z
AND(x', y, z) → x'yz
AND(x, y') → xy'
OR(x'y'z, x'yz, xy') → F_2
= 3 AND + 1 OR + 2 NOT (3 levels if NOTs counted)

(b) Simplified — $F_2 = x'z + xy'$:
NOT(x) → x'   NOT(y) → y'
AND(x', z) → x'z
AND(x, y') → xy'
OR(x'z, xy') → F_2
= 2 AND + 1 OR + 2 NOT (TRUE 2-level SOP)

Section 3

De Morgan's Theorem & Complements

Find function complements; verify with truth tables

CW 4 — Find the Complement of Each Function

Section 3: De Morgan's Theorem

Apply De Morgan's theorems repeatedly to find F' for each function. Simplify your result to a minimal form.

1

Find F_1'

$$F_1 = x'yz' + x'y'z$$

Recall: (A+B)' = A'B' and (AB)' = A'+B'

3

Find F_3' — two methods

$$F_3 = (A+B)(A'+C)$$

Recall: (A+B)' = A'B' and (AB)' = A'+B'

2

Find F_2'

$$F_2 = x(y'z' + yz)$$

Recall: (A+B)' = A'B' and (AB)' = A'+B'

4

Apply Consensus theorem first

$$F_4 = AB + A'CD + BCD$$

Recall: (A+B)' = A'B' and (AB)' = A'+B'

CW 4 — Solutions: Function Complements

De Morgan applied step-by-step

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1

$F_1' = (x'yz'+x'y'z)' = (x'yz')'\cdot(x'y'z)'$
$\quad = (x+y'+z)(x+y+z')$

Note: (Sum of products → product of sums via dual DeMorgan)

3

Method 1 (DeMorgan directly):
$F_3' = [(A+B)(A'+C)]' = (A+B)' + (A'+C)'$
$\quad = A'B' + A(C')$
$F_3' = A'B' + AC'$

Method 2 (Dual + complement literals):
Dual of $F_3 = (AB) + (A'C)$
Complement literals: $(A'B') + (AC') = F_3'$ ✓

Note: (Both methods yield the same result)

2

$F_2' = [x(y'z'+yz)]' = x' + (y'z'+yz)'$
$\quad = x' + (y'z')' \cdot (yz)'$
$\quad = x' + (y+z)(y'+z')$

Note: (Distribute outer: x' + inner complement)

4

First apply Consensus T9: BCD is consensus of AB & A'CD
$F_4 = AB + A'CD$ (BCD removed)

Now find complement:
$F_4' = (AB + A'CD)' = (AB)'\cdot(A'CD)'$
$\quad = (A'+B')\cdot(A+C'+D')$

Note: (Simplify first using Consensus, then complement)

Section 4

Algebraic Simplification

Apply theorems to minimize literals and gate count

CW 5 — Simplify to Minimum Literals (Example 2.1)

Section 4: Algebraic Manipulation

Simplify each expression to the minimum number of literals. Cite the theorem used at each step.

1. **Simplify:** $x(x' + y)$

2. **Simplify:** $x + x'y$

3. **Simplify:** $(x + y)(x + y')$

4. **Simplify:** $xy + x'z + yz$

5. **Simplify:** $AB + A'C + BC$

6. **Simplify:** $(x + y)(x' + z)(y + z)$

CW 5 — Solutions: Simplify to Minimum Literals

Mano Example 2.1 — all 5 results plus Consensus

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1 $x(x' + y)$

$= xx' + xy$ [P5a]
$= 0 + xy$ [P6b]
$= xy$ [P2a]

2 $x + x'y$

$= (x+x')(x+y)$ [P5b]
$= 1\cdot(x+y)$ [P6a]
$= x + y$ [P2b]

3 $(x+y)(x+y')$

$= x + yy'$ [P5b]
$= x + 0$ [P6b]
$= x$ [P2a]

4 $xy + x'z + yz$

$= xy + x'z + yz(x+x')$ [P6a]
$= xy+x'z+xyz+x'yz$
$= xy(1+z)+x'z(1+y)$
$= xy + x'z$ [T2a]

5 $AB + A'C + BC$

BC is consensus of AB & A'C
[T9 Consensus theorem]
$= AB + A'C$

6 $(x+y)(x'+z)(y+z)$

Dual of Ex.4: $AB+A'C+BC$
→ dual is $(A+B)(A'+C)(B+C)$
$= (A+B)(A'+C)$ [Consensus]
So: $(x+y)(x'+z)(y+z)$
$= (x+y)(x'+z)$

CW 6 — NAND Network: $f(X,Y,Z) = \Sigma m(0,3,4,5,7)$

Section 4: NAND Implementation (Slide 41 Example)

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Given $f(X,Y,Z) = \Sigma m(0,3,4,5,7)$. (a) Expand to canonical SOP. (b) Simplify using adjacency. (c) Convert to NAND-NAND form and draw.

Step 1

Expand minterms

m_0 = X'Y'Z' (000)
m_3 = X'YZ (011)
m_4 = XY'Z' (100)
m_5 = XY'Z (101)
m_7 = XYZ (111)

f = X'Y'Z' + X'YZ + XY'Z' + XY'Z + XYZ

Step 2

Simplify (adjacency T6)

XY'Z' + XY'Z = XY' (z cancels)
X'Y'Z' + XY'Z' = Y'Z' (x cancels)
X'YZ + XYZ = YZ (x cancels)

f = XY' + Y'Z' + YZ

Step 3

NAND-NAND form

f = XY' + Y'Z' + YZ
= ((XY')'·(Y'Z')'·(YZ)')'

Circuit: Three 2-input NAND gates
feeding one 3-input NAND gate

= NAND(NAND(X,Y'), NAND(Y',Z'), NAND(Y,Z))

Section 5

Minterms, Maxterms & Canonical Forms

SOM · POM · Example 2.4 · Example 2.5 · Conversions

CW 7 — Identify Minterms & Maxterms

Section 5: Minterms & Maxterms

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For each entry: (a) write the minterm m_i and maxterm M_i for the given variable combination; (b) give the complementary relationship.

i	x	y	z	Minterm m_i	Maxterm M_i	Relation
0	0	0	0	x'y'z'	x+y+z	$(m_0)' = M_0$
3	0	1	1	x'yz	x+y'+z'	$(m_3)' = M_3$
5	1	0	1	xy'z	x'+y+z'	$(m_5)' = M_5$
6	1	1	0	xyz'	x'+y'+z	$(m_6)' = M_6$
7	1	1	1	xyz	x'+y'+z'	$(m_7)' = M_7$
?	0	1	0	**? — write it**	**? — write it**	**? relation**
?	1	0	0	**? — write it**	**? — write it**	**? relation**

✓ Solutions for rows 6 & 7

Row 6: x=0, y=1, z=0 → index = $010_2 = 2$
m_2 = x'yz' (0→complement, 1→normal)
M_2 = x+y'+z (0→normal, 1→complement)
$(m_2)' = M_2$ ✓

Row 7: x=1, y=0, z=0 → index = $100_2 = 4$
m_4 = xy'z'
M_4 = x'+y+z
$(m_4)' = M_4$ ✓

Index rule summary:
Minterm: 1→uncomplemented, 0→complemented
Maxterm: 0→uncomplemented, 1→complemented

CW 8 — Canonical SOM & POM: Examples 2.4 & 2.5 (Mano)

Section 5: Sum-of-Minterms · Product-of-Maxterms

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Express each function in both canonical SOM and canonical POM. Show the algebraic expansion and the final index notation.

1 F(A,B,C) = A + B'C (Example 2.4)

→ Express as Σm(...) and ΠM(...)

2 F(x,y,z) = xy + x'z (Example 2.5)

→ Express as ΠM(...) and Σm(...)

✓ Solution — Example 2.4

Expand: F = A(B+B') + B'C
= AB + AB' + B'C
= AB(C+C') + AB'(C+C') + (A+A')B'C
= ABC + ABC' + AB'C + AB'C' + A'B'C

Minterms where F=1:
A'B'C=m_1, AB'C'=m_4, AB'C=m_5, ABC'=m_6, ABC=m_7

∴ F(A,B,C) = Σm(1, 4, 5, 6, 7)

POM indices = complement of SOM indices:
{0,1,2,3,4,5,6,7} \ {1,4,5,6,7} = {0,2,3}
∴ F(A,B,C) = ΠM(0, 2, 3)

✓ Solution — Example 2.5

Expand: F = xy + x'z = (xy+x')(xy+z)
= (x+x')(y+x')(x+z)(y+z)
= 1·(x'+y)(x+z)(y+z)

Expand each sum term to 3 variables:
(x'+y) = (x'+y+z)(x'+y+z')
(x+z) = (x+y+z)(x+y'+z)
(y+z) = (x+y+z)(x'+y+z)

Maxterms where F=0: M_0,M_2,M_4,M_5

∴ F(x,y,z) = ΠM(0, 2, 4, 5)

SOM indices = complement of POM:
∴ F(x,y,z) = Σm(1, 3, 6, 7)

CW 9 — SOM ↔ POM Conversion & Truth Table

Section 5: Canonical Form Conversions

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Given $F(x,y,z) = \Sigma m(2,3,5,7)$. (a) Write the POM form. (b) Complete the truth table. (c) Find $F'(x,y,z)$.

i	x	y	z	F	Minterm/Maxterm
0	0	0	0	0	$M_0 = x+y+z$
1	0	0	1	0	$M_1 = x+y+z'$
2	0	1	0	**1**	$m_2 = x'yz'$
3	0	1	1	**1**	$m_3 = x'yz$
4	1	0	0	0	$M_4 = x'+y+z$
5	1	0	1	**1**	$m_5 = xy'z$
6	1	1	0	0	$M_6 = x'+y'+z$
7	1	1	1	**1**	$m_7 = xyz$

✓ Answers

(a) POM form:
Maxterm indices = $\{0,1,\ldots,7\} \setminus \{2,3,5,7\} = \{0,1,4,6\}$
$F(x,y,z) = \Pi M(0,1,4,6)$
$= (x+y+z)(x+y+z')(x'+y+z)(x'+y'+z)$

(b) Truth table: F=1 for rows 2,3,5,7 (shaded green)
F=0 for rows 0,1,4,6

(c) Complement F':
$F = \Sigma m(2,3,5,7) \rightarrow F' = \Sigma m(0,1,4,6)$
OR equivalently: $F' = \Pi M(2,3,5,7)$

Key rule: SOM and POM index sets are complementary.
To convert: list the MISSING indices.

Section 6

Standard Forms & Gate Implementation

SOP · POS · NAND-NAND · Circuit Analysis & Synthesis

CW 10 — Circuit Analysis: Majority & XOR-AND Circuit

Section 6 / Section 8: Algebraic Analysis

Derive the Boolean expression for each circuit, simplify, identify the function name, and find the canonical SOM.

1 Circuit 1 — Three AND gates → OR gate (Slide 55)

$g_1 = xy,\ \ g_2 = yz,\ \ g_3 = xz;\ \ \ f = g_1 + g_2 + g_3$

→ *Simplify and identify the circuit function.*

2 Circuit 2 — XOR-AND circuit (Slide 45 Example)

$g_1 = a \oplus b,\ \ g_2 = g_1 \cdot c,\ \ g_3 = ab;\ \ \ f = g_2 + g_3$

→ *Simplify fully and name the function.*

✓ Solution 1

f = xy + yz + xz

This is the MAJORITY function:
F=1 when at least 2 of the 3 inputs are 1

Truth table check:
xy+yz+xz = 1 only for (1,1,0),(1,0,1),(0,1,1),(1,1,1)
i.e., rows 3, 5, 6, 7

Canonical SOM: Σm(3, 5, 6, 7)
Canonical POM: ΠM(0, 1, 2, 4)

✓ Solution 2

f = (a⊕b)·c + ab
= (a'b+ab')c + ab [expand XOR]
= a'bc + ab'c + ab [distribute]
= a'bc + a(b'c+b) [factor a]
= a'bc + a(b+c) [T5: b'c+b=b+c]
= a'bc + ab + ac

= ab + ac + bc (MAJORITY function in a,b,c!)

Both circuits implement the same Majority function.
Canonical SOM: Σm(3, 5, 6, 7)

CW 11 — Circuit Synthesis: Design from Word Description

Section 6: Circuit Synthesis (Slide 42 Examples)

Convert each word description into Boolean equations, then express in standard SOP and draw the gate diagram.

1 Burglar Alarm

A = secret switch B = safe in normal position
C = clock (10:00–14:00) D = closet door closed
Alarm triggers if:

- Safe moved with secret switch ON (A·B')
- Closet opened after hours (D'·C')
- Closet opened, switch open (D'·A')

✓ Solution 1

Alarm = AB' + D'C' + D'A'

Standard SOP (already minimal):
3 product terms, 6 literals

Gate diagram:
NOT(B)→B', NOT(C)→C', NOT(A)→A', NOT(D)→D'
AND(A,B') → AB'
AND(D',C') → D'C'
AND(D',A') → D'A'
OR(AB', D'C', D'A') → Alarm

= 4 NOT + 3 AND + 1 OR (standard 2-level SOP)

2 Family Voter Circuit

A=Dad, B=Mom, C=Joe, D=Sue (0=hamburger, 1=chicken)
Rules: (1) Majority vote wins;
(2) If Mom and Dad agree, they override kids;
(3) Otherwise majority of all four wins.

✓ Solution 2

F = AB + ACD + BCD

Explanation:
AB → Mom and Dad both want chicken (override)
ACD → Dad + Joe + Sue want chicken (majority)
BCD → Mom + Joe + Sue want chicken (majority)

Gate diagram:
AND(A,B) → AB
AND(A,C,D) → ACD
AND(B,C,D) → BCD
OR(AB,ACD,BCD) → F

= 3 AND + 1 OR (clean 2-level SOP)

Quick Reference — Boolean Algebra Classwork Toolkit

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Key Postulates (P1–P6)

P2: $x+0=x$, $x\cdot 1=x$
P3: $x+y=y+x$ (commutative)
P4: $x+(y+z)=(x+y)+z$
P5: $x\cdot(y+z)=xy+xz$
$x+(y\cdot z)=(x+y)\cdot(x+z)$
P6: $x+x'=1$, $x\cdot x'=0$

Key Theorems (T1–T9)

T1: $x+x=x$, $x\cdot x=x$ (idempotent)
T2: $x+1=1$, $x\cdot 0=0$ (null)
T3: $(x')'=x$ (involution)
T5: $x+x'y=x+y$ (simplification)
T6: $AB+AB'=A$ (adjacency)
T8: $(x+y)'=x'y'$ (DeMorgan)
T9: $AB+A'C+BC=AB+A'C$

Canonical Form Rules

SOM: OR of minterms (F=1 rows)
POM: AND of maxterms (F=0 rows)
Index rule — minterm:
1→uncomplemented, 0→complemented
Index rule — maxterm:
0→uncomplemented, 1→complemented
$(m_i)' = M_i$ for same index i

Simplification Checklist

1. Adjacency: $AB+AB'=A$
2. Absorption: $A+AB=A$
3. Simplification: $A+A'B=A+B$
4. De Morgan complement
5. Consensus removal:
   $AB+A'C+BC \rightarrow AB+A'C$
6. Distribution/factoring
7. Verify with truth table

SOM↔POM Conversion

$\Sigma m(S) = \Pi M(\text{complement of } S)$
Complement of S: all indices in
$\{0,\ldots,2^n-1\}$ NOT in S

Example (3 vars, n=3):
$\Sigma m(1,4,7) = \Pi M(0,2,3,5,6)$

F' has same indices, different Σ/Π:
$F=\Sigma m(S) \rightarrow F'=\Sigma m(\text{complement of } S)$
$F=\Sigma m(S) \rightarrow F'=\Pi M(S)$

Gate Implementation

2-level SOP: AND→OR
2-level POS: OR→AND
NAND-NAND ≡ AND-OR
NOR-NOR ≡ OR-AND

Building with NAND:
NOT: A NAND A
AND: NAND then NAND
OR: NOT(A) NAND NOT(B)

Always: 2-level = 2 gate delays